

Mount all the components as shown in the PCB layout. The ATmega2560 used in the prototype comes in a 100-pin, surface-mount TQFP package. By using surface-mount devices (SMDs) in your project, you don't have to drill holes in the PCB and the board size will be much smaller. But it may be difficult to handle and solder SMDs without an SMD soldering station. However, with a normal soldering iron, tweezers and some basic knowledge, you can solder SMD components without much problem. You need good eyes, a steady hand and a soldering iron with a small, clean tip.

Make sure that the PCB is clean.

You may use copper polish material to remove all kinds of oxidation and acetone (nail polish remover) to remove all kinds of contaminations on the PCB. On the PCB, figure out the correct location for placing the SMD. Fixate the SMD on the top-right corner and the bottom-left corner. Carefully solder SMD pins one by one. SMD components are very sensitive to heat, so allow your SMD to cool down after every step.

For more tips on soldering SMD components, you may refer to <http://hem.passagen.se/communication/pcbssmd.html> and www.engadget.com/2006/03/07/how-to-make-a-sur-

face-mount-soldering-iron/ websites.

Now connect a 9V DC source to the circuit and switch on the power supply. LED1 should glow to indicate the presence of power supply in the circuit. Now run the HyperTerminal or Terminal software and send the text file (.txt) to the LCD from your computer.

EFY note. The source code of this article has been included in this month's EFY-CD and is also available on efymag.com website.

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I2C_EEPROM.C

```
Source code : I2C_EEPROM.c
/* I2C Implementation routine for Serial EEPROM chip AT24C32 using ATmega2560 and 16Mhz External Crystal */
#include <iavr.h>
#include <intrinsics.h>

#define START 0x08
#define REP_START 0x10
#define MT_SLA_ACK 0x18
#define MT_DATA_ACK 0x28
#define MR_SLA_ACK 0x40
#define MR_SLA_NACK 0x48
#define MR_DATA_NACK 0x58
#define STATUS_MASK (1<<TWS7 | 1<<TWS6)

1<<TWS5 | 1<<TWS4 | 1<<TWS3)
#define STATUS (TWSR & STATUS_MASK)

#define SETBIT(ADDRESS, BIT) (ADDRESS |= (1 << BIT))
#define CLEARBIT(ADDRESS, BIT) (ADDRESS &= ~(1 << BIT))
#define CHECKBIT(ADDRESS, BIT) (ADDRESS & (1 << BIT))
```

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